

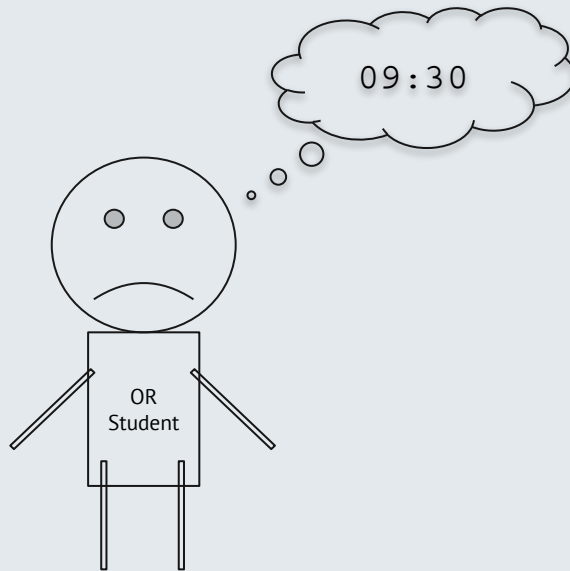


Optimizing Passenger Waiting Time in Taipei Metro

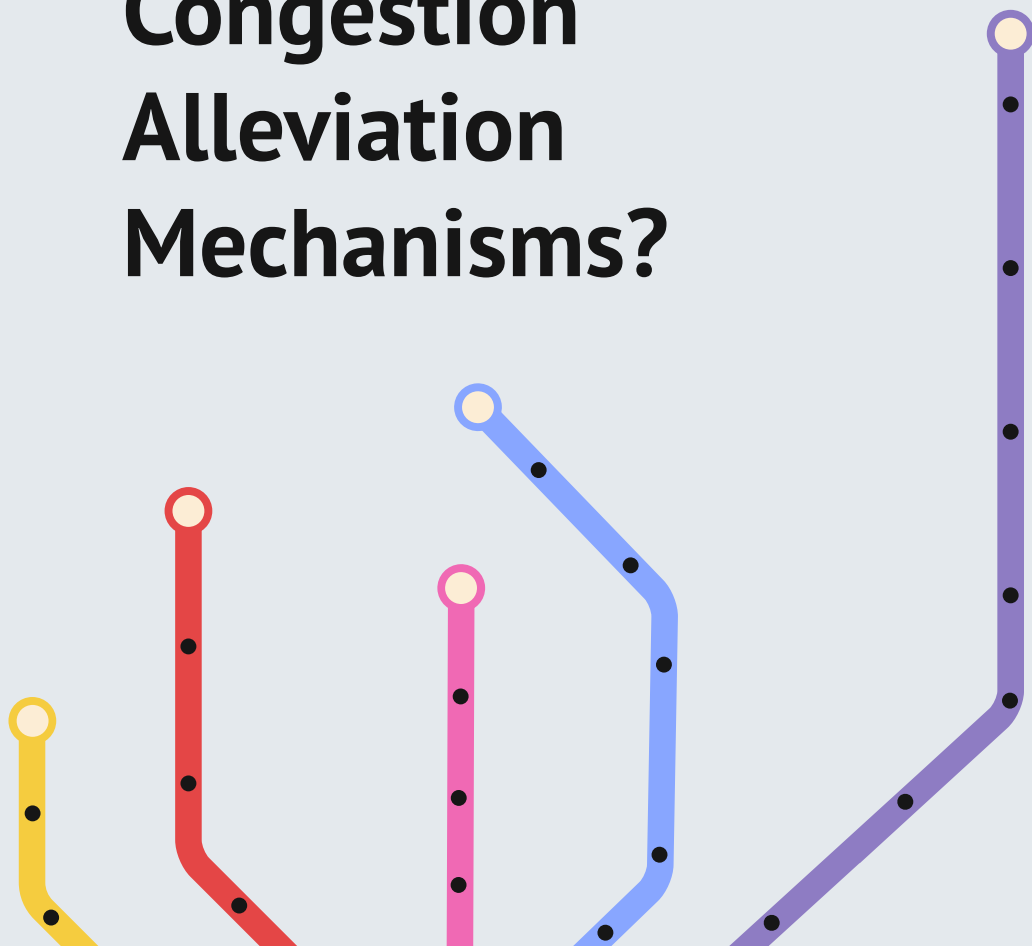
Operations Research, Spring 2024 (112-2)
Final Project - Group G

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A Small Scenario...



Congestion Alleviation Mechanisms?

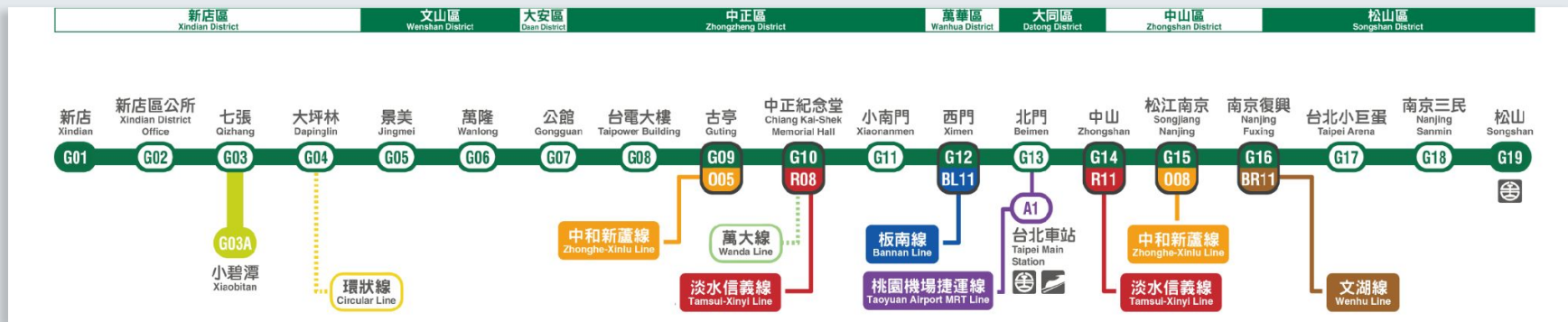


- Increasing the **frequency** of train departures (reducing the headway).
- Adding short-distance “**shuttle**” train.

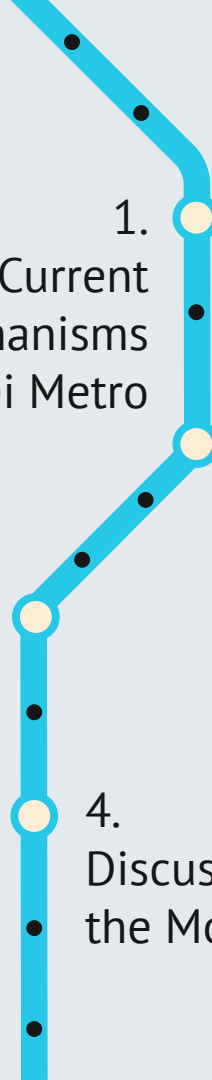
Our Objective

To optimize passenger waiting time, focusing on:

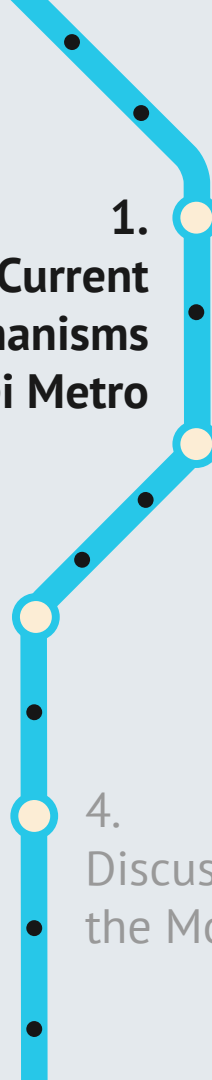
-  松山新店線
Songshan-Xindian “**Green**” Line,
- weekday **peak** hours: 07:00-09:00, 17:00-19:30.



Outline

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1. Current Congestion Alleviation Mechanisms of the Taipei Metro
 2. Our Objectives and Decision Factors
 3. A Brief Overview of the Model and Important Constraints
 4. Discussion of the Model Results

Outline

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1. **Current Congestion Alleviation Mechanisms of the Taipei Metro**
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Current Mechanisms

02.
Increasing
the **frequency**
of train departures
(reducing the headway).

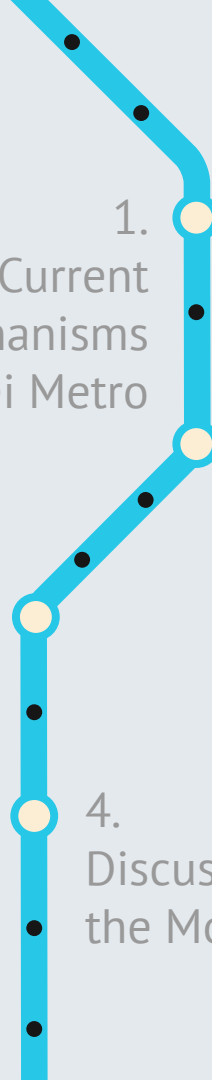
Period/Type	Full-Distance Train	Shuttle
Peak Hours	every 4 - 6 minutes	every 3 minutes
Off-Peak Hours	every 6 - 8 minutes	N/A

01.

Adding short-distance “**shuttle**” train from Songshan to Taipower Building.



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Dataset and Model Scenario

日期	時段	進站	出站	人次
2024-04-29	18	中山	西門	238
2024-04-29	18	中山	善導寺	21
2024-04-29	18	中山	忠孝新生	22
2024-04-29	18	中山	忠孝敦化	40
2024-04-29	18	中山	國父紀念館	19
2024-04-29	18	中山	市政府	40

*“Hourly **entry** and **exit** counts
for each station (raw data).”*

日期	時段	站	進站人次	出站人次
2024-04-29	18	中山	4421	5525

*“Hourly **entry** and **exit** counts
for each station (processed data).”*

- Substituting the exact total waiting time with the **sum of average waiting times per minute**.
- Do not consider the distance differences between stations.

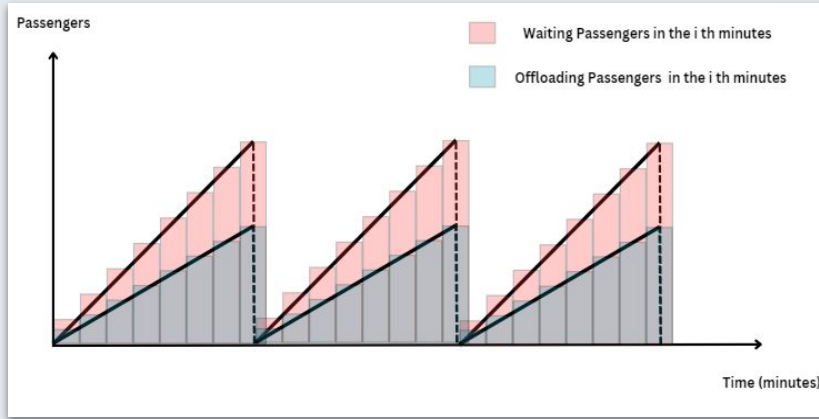
Analysis of Waiting Time

- Passengers are **evenly distributed** across the 60 minutes.
 - At the specified hour, the number of passengers **entering** (**exiting**) station per minute, is the same.
 - Dividing the hourly passengers **entry** (**exit**) count from our dataset by 60 minutes.
- During the interval between trains, **boarding** (**offloading**) passengers **increases gradually at this per-minute rate**.
 - The **basic waiting time** at each MRT station.
 - The next train may not accommodate all waiting passengers: resulting in **additional waiting time**.

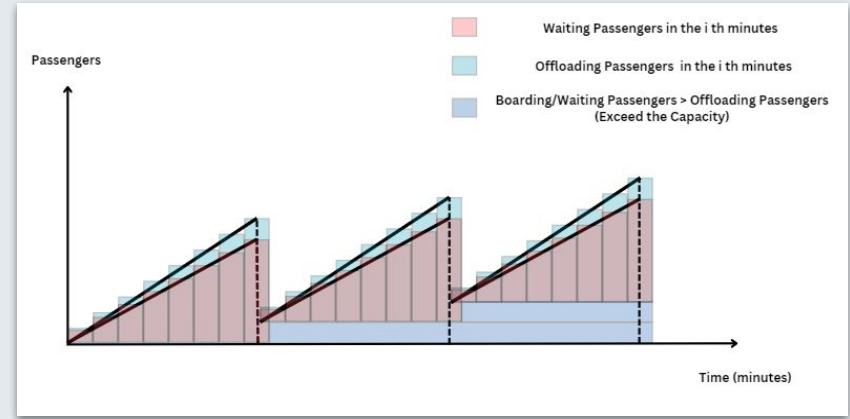
日期	時段	站	進站人次	出站人次
2024-04-29	18	中山	4421	5525

*“Hourly **entry** and **exit** counts
for each station (processed data).”*

Analysis of Waiting Time



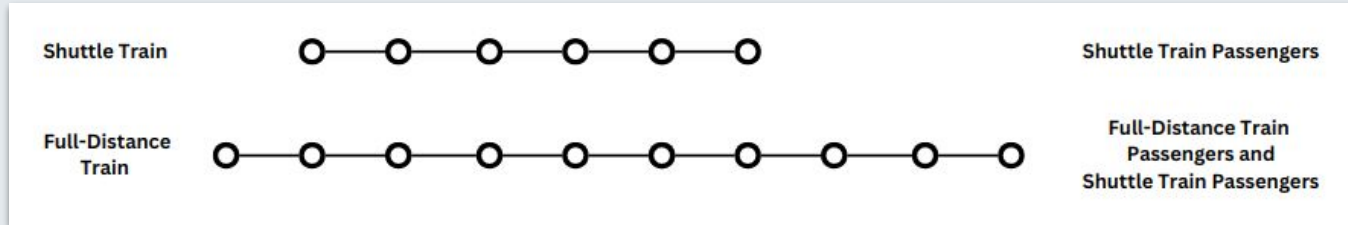
*“**Boarding** passengers $\leq$ **offloading** passengers
(**remain within** the capacity, **basic** waiting time).”*



*“**Boarding** passengers > **offloading** passengers
(**exceed** the capacity, **basic + additional** waiting time).”*

- The waiting passengers between each train at this station, at the specified hour, will only continue to “**exceed the capacity**” or “**remain within the capacity**”.

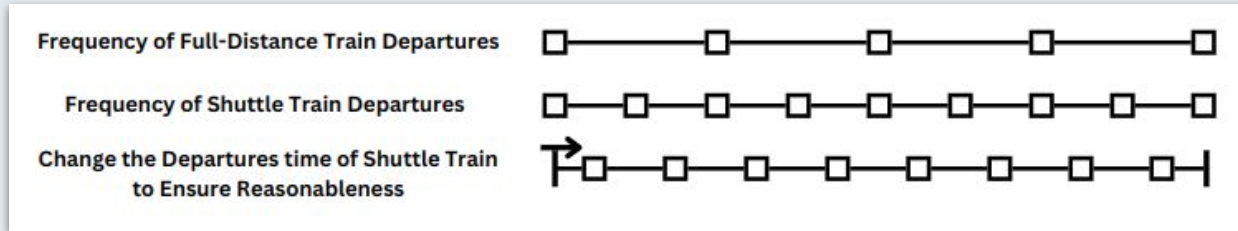
Shuttle and Full-Distance Trains



"Shuttle and full-distance trains."

- Passengers of full-distance trains are not allowed to transfer (to take shuttle first).
- Passengers can be categorized based on their destination:
 - (1) prioritizing passengers **"eligible to take shuttle"**, to **board shuttle**,
 - (2) the passengers who **can only take full-distance** trains,
 - (3) the passengers who cannot board shuttle **due to overcrowding**, consequently **board full-distance** trains.

Frequency of Train Departures



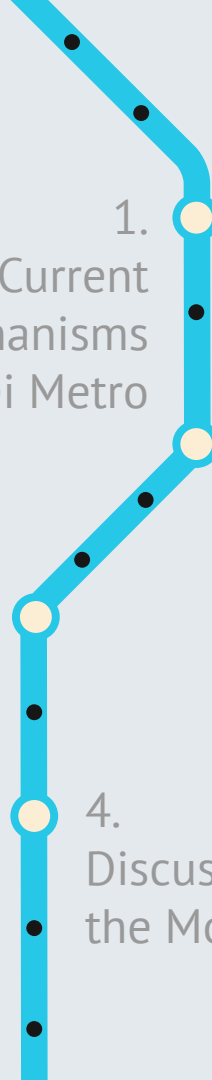
"Shuttle and full-distance trains."

- The departure intervals of shuttle and full-distance trains must be in a certain proportion.
- The **full-distance** train must depart **at the "midpoint" between** the departures of two **shuttles**.
 - Minimizes the accumulation of waiting passengers in any given time interval.

Summary of Objectives

- Aim to **minimize the waiting time for all passengers**, by **determining**:
 - a. the **frequencies** of "full-distance" and "shuttle" trains,
 - b. **selecting the two stations** as the start and end points for the **shuttle**.
- Use the model and the dataset to determine the optimal values for these variables:
 - a. the frequency of shuttle,
 - b. the frequency of full-distance trains,
 - c. the number of shuttle,
 - d. the number of full-distance trains,
 - e. the starting and ending stations for shuttle.

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Important Constraints

- Number of full-distance + shuttle trains
≤ Green Line trains Taipei Metro can provide.
- Frequency of train departures must not be too small to prevent
 - a. train collisions,
 - b. or insufficient trains available.
- Full-distance and shuttle trains departure times should be staggered according to a certain ratio.

$$\begin{aligned} \text{s.t.} \quad & \frac{60}{k^s} \geq \frac{2(m-1)T}{r^s} \\ & \frac{60}{k^l} \geq \frac{2(n-1)T}{r^l} \\ & r^l + r^s \leq R \\ & \alpha k^l = k^s, \alpha \in \mathbb{N} \\ & \frac{60}{k^l} \geq Z, \frac{60}{k^s} \geq Z \end{aligned}$$

"Important Constraints."

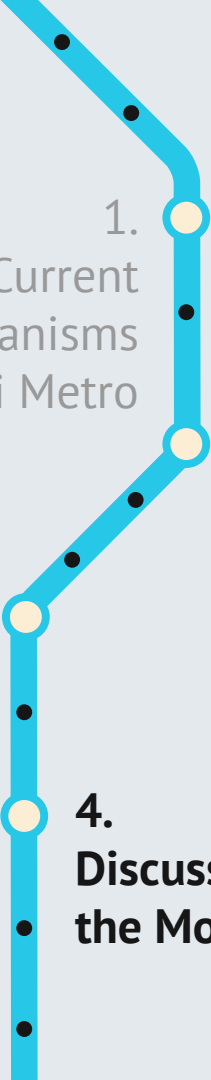
Objective Function

- We categorized passengers into three types (page 12).
- Sum the waiting times for each of the three types of passengers separately.

$$\min \quad 30 \sum_{i=1}^n \left\{ \frac{b_i}{k^s} [I_i^s + (k^s - 1)p_i^s] + \frac{1}{k^l} [I_i^l + (k^l - 1)p_i^l] \right\}$$

"Objective Function."

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Results

G R

G O

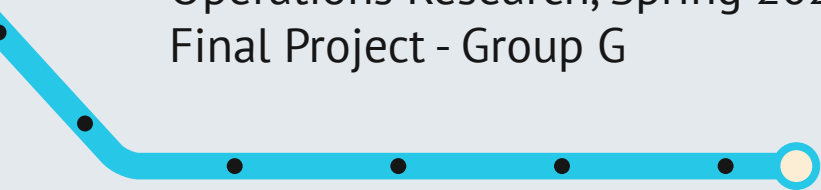
- **Chiang Kai-Shek Memorial Hall** Station (Nanmen) to **Guting** Station.
- Both full-distance and shuttle trains having a departure frequency of 10 trains/hour.

G BR

G O

- **Nanjing Fuxing** Station to **Songjiang Nanjing** Station having **no available capacity** on the trains after departure → **consider dual-segment?**
- Would **increase the cost** of constructing the "pocket tracks/reversing siding" needed for train turnarounds.





Thanks!

Operations Research, Spring 2024 (112-2)
Final Project - Group G

